Chapter 5. Partial Identification and Soft Interventions

Natural bounds, the instrumental-variable linear program, stochastic interventions and sigma-calculus

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5.1 Partial identification

Motivation. When $P(y \mid \mathrm{do}(x))$ is not identifiable, we seek the tightest interval $[a(y;x), b(y;x)] \subseteq [0,1]$ consistent with the data and the graph.

5.1.1 The partial identification problem

Motivation. Extend the yes/no identification question to bounds.

When $P(y \mid \mathrm{do}(x))$ is not determined by $P(\mathbf{v})$ and $\mathcal{G}$, we ask for a non-trivial bound

$$P(y \mid \mathrm{do}(x)) \in [a(y;x),\ b(y;x)] \subseteq [0,1].$$

Definition (Partial identifiability)

The effect of X on Y is partially identifiable given $\mathcal{G}$ if $P(\mathbf{V})$ determines a bound $P(y \mid \mathrm{do}(x)) \in [a,b]$ that is a proper subset of $[0,1]$.

Identifiable $\Rightarrow$ partially identifiable ($a = b$), not conversely.

5.1.2 Natural bounds

Motivation. The simplest bounds hold for any confounded pair and cannot be improved without more structure.

Theorem 5.1 (Natural bounds; Manski 1990, Robins 1989)

For an SCM compatible with $X \to Y$, $X \leftrightarrow Y$ (latent U), given $P(x,y)$,

$$a(y;x) = P(y,x) \ \le\ P(y \mid \mathrm{do}(x)) \ \le\ P(y,x) + 1 - P(x) = b(y;x).$$

Proof [Supplement]. With $X \leftarrow f_X(u)$, $Y \leftarrow f_Y(x,u)$, the truncated product gives

$$\begin{aligned} P(y \mid \mathrm{do}(x)) &= \sum_u P(y \mid x,u)P(u) = \sum_u P(y \mid x,u)\big(P(u,x) + P(u) - P(u,x)\big) \\ &= \sum_u P(y,x,u) + \sum_u P(y \mid x,u)\big(P(u) - P(u,x)\big) \\ &= P(y,x) + \sum_u P(y \mid x,u)\big(P(u) - P(u,x)\big). \end{aligned}$$

As $0 \le P(y \mid x,u) \le 1$ and $P(u) - P(u,x) \ge 0$, the last sum lies in $[0, \sum_u (P(u) - P(u,x))] = [0, 1 - P(x)]$. ■

The width $1 - P(x)$ is the mass of units not receiving x; equivalently $b(y;x) = 1 - P(x,y')$.

Example 5.1

$X, Y, U \in \{0,1\}$, $X \leftarrow U$, $Y \leftarrow X \oplus U$, $P(U=0) = 0.9$. Observationally $Y = 0$: $P(X=0, Y=0) = 0.9$, $P(X=1, Y=0) = 0.1$. Under $\mathrm{do}(X=x)$, $Y = x \oplus U$:

	$P(Y=1 \mid \mathrm{do}(x))$	$a(Y=1;x)$	$b(Y=1;x)$
$x=0$	0.1	0	0.1
$x=1$	0.9	0	0.9

The true effects lie on the upper bounds.

5.1.2.1 Tightness

Theorem 5.2

For binary X, Y and any $P(x, y)$ there exist SCMs $\mathcal{M}^{(1)}, \mathcal{M}^{(2)}$ with $P^{(1)}(x, y) = P^{(2)}(x, y) = P(x, y)$ and $P^{(1)}(y \mid \mathrm{do}(x)) = a(y; x)$, $P^{(2)}(y \mid \mathrm{do}(x)) = b(y; x)$.

Proof [Supplement]. Take any compatible $\mathcal{M}$ ($X \leftarrow f_X(u)$, $Y \leftarrow f_Y(x, u)$) inducing P, and define, with the same $P(U)$ and $X \leftarrow f_X(u)$,

$$\mathcal{F}^{(1)}: \; Y \leftarrow \begin{cases} f_Y(x,u) & x = f_X(u) \\ 0 & \text{otherwise} \end{cases}, \qquad \mathcal{F}^{(2)}: \; Y \leftarrow \begin{cases} f_Y(x,u) & x = f_X(u) \\ 1 & \text{otherwise} \end{cases}.$$

Without intervention $X = f_X(U)$, so $Y = f_Y(X, U)$ and $P^{(i)}(x, y) = \sum_u P(x \mid u)P(y \mid x, u)P(u) = P(y, x)$. Under $\mathrm{do}(X = x)$,

$$\begin{aligned} P^{(i)}(Y = 1 \mid \mathrm{do}(x)) &= P(Y = 1 \mid x, f_X(U) = x)P(f_X(U) = x) \\ &\quad + P^{(i)}(Y = 1 \mid x, f_X(U) \neq x)P(f_X(U) \neq x) \\ &= P(x, Y = 1) + \mathbb{1}[i = 2](1 - P(x)). \end{aligned}$$

So $\mathcal{M}^{(1)}$ attains a, $\mathcal{M}^{(2)}$ attains b ($Y = 0$ is symmetric). ■

5.1.2.2 Limitations

Natural bounds are tight for the binary bow graph but not when more structure (e.g. an instrument Z) is available; canonical type models and linear programming exploit it.

5.1.3 Canonical type models and LP bounds

Motivation. Replace unknown functional forms by finitely many response types, turning bounding into a linear program.

5.1.3.1 Experiments with non-compliance

Assigned treatment Z (randomized instrument), received treatment X, outcome Y, latent U: $Z \to X \to Y$, $X \leftrightarrow Y$. Given $P(x, y \mid z)$, bound $P(y \mid \mathrm{do}(x))$.

Theorem 5.3 (Natural bounds in the IV model)

$$a(y; x) = \max_z P(y, x \mid z), \qquad b(y; x) = \min_z \left[P(y, x \mid z) + 1 - P(x \mid z)\right].$$

Proof [Supplement]. Fix z. Natural bounds in $\mathcal{M}_z$: $P(y, x \mid \mathrm{do}(z)) \leq P(y \mid \mathrm{do}(x, z)) \leq P(y, x \mid \mathrm{do}(z)) + 1 - P(x \mid \mathrm{do}(z))$. Exclusion gives $P(y \mid \mathrm{do}(x, z)) = P(y \mid \mathrm{do}(x))$, randomization $P(\cdot \mid \mathrm{do}(z)) = P(\cdot \mid z)$. Take $\max_z$ of the lower and $\min_z$ of the upper bounds. ■

They are not tight in IV models.

5.1.3.2 Optimization perspective

With $\mathcal{M}$ the SCMs compatible with $Z \leftarrow f_Z(U_Z)$, $X \leftarrow f_X(Z,U)$, $Y \leftarrow f_Y(X,U)$, the sharp bounds are

$$a(y;x) = \min_{M\in\mathcal{M}:\, P_M(x,y|z)=P(x,y|z)} P_M(y \mid \mathrm{do}(x)), \qquad b(y;x) = \max_{M\in\mathcal{M}:\, P_M(x,y|z)=P(x,y|z)} P_M(y \mid \mathrm{do}(x)),$$

hard to solve directly since $\mathcal{F}$ and $P(\mathbf{U})$ are unspecified.

5.1.3.3 U as a selector and response variables

For fixed $U = u$, $f_X(\cdot, u)$ is a function $Z \to X$: U *selects* the mapping, so $\mathfrak{X}_U$ is partitioned by the induced function. Binary Z, X give $|\mathfrak{X}_X|^{|\mathfrak{X}_Z|} = 4$ cells:

$$\{u : f_X(0,u) = 0,\ f_X(1,u) = 0\} \quad (X = 0), \qquad \{u : f_X(0,u) = 0,\ f_X(1,u) = 1\} \quad (X = Z),$$
$$\{u : f_X(0,u) = 1,\ f_X(1,u) = 0\} \quad (X = 1 - Z), \qquad \{u : f_X(0,u) = 1,\ f_X(1,u) = 1\} \quad (X = 1)$$

(ternary Z: 2^3). A *response variable* indexes the partition:

$$R_X = 0\text{: } X = 0 \text{ (never-taker)} \quad R_X = 1\text{: } X = Z \text{ (complier)} \quad R_X = 2\text{: } X = 1 - Z \text{ (defier)}$$
$$R_X = 3\text{: } X = 1 \text{ (always-taker)}$$

Likewise $R_Y \in \{0,1,2,3\}$: never-recover ($Y = 0$), helped ($Y = X$), hurt ($Y = 1-X$), always-recover ($Y = 1$). The 16 cells have probabilities

$$q_{j,k} = P(R_X = j, R_Y = k), \qquad \sum_{j=0}^{3}\sum_{k=0}^{3} q_{j,k} = 1.$$

Definition (Canonical type model)

An SCM with $X, Y, Z \in \{0,1\}$ and $\mathbf{U} = \{R_X, R_Y, R_Z\}$, $R_X, R_Y \in \{0,1,2,3\}$, $R_Z \in \{0,1\}$:

$$Z \leftarrow r_Z, \qquad X \leftarrow f_X(z, r_X) = \begin{cases} 0 & r_X = 0 \\ z & r_X = 1 \\ 1 - z & r_X = 2 \\ 1 & r_X = 3 \end{cases}, \qquad Y \leftarrow f_Y(x, r_Y) = \begin{cases} 0 & r_Y = 0 \\ x & r_Y = 1 \\ 1 - x & r_Y = 2 \\ 1 & r_Y = 3 \end{cases}.$$

Theorem 5.4 (Correspondence; Balke and Pearl 1997)

For any IV model $\mathcal{M}_1$ there is a canonical type model $\mathcal{M}_2$ with $P^{\mathcal{M}_1}(x, y \mid z) = P^{\mathcal{M}_2}(x, y \mid z)$ and $P^{\mathcal{M}_1}(y \mid \mathrm{do}(x)) = P^{\mathcal{M}_2}(y \mid \mathrm{do}(x))$.

Proof [Supplement]. Partition $\mathfrak{X}_U$ by $u \mapsto (x(\cdot), y(\cdot))$, $x(z) = f_X(z,u)$, $y(x) = f_Y(x,u)$: $4\times4 = 16$ cells. Let $\mathcal{M}_2$ have $R = (R_X, R_Y) = (j,k)$ with probability $q_{j,k}$, the $P(U)$-mass of cell (j,k), and $X = \mathrm{response}_j(Z)$, $Y = \mathrm{response}_k(X)$. Both $P^{\mathcal{M}_1}(x, y \mid z)$ (mass of $\{u : f_X(z,u) = x, f_Y(x,u) = y\}$) and $P^{\mathcal{M}_1}(y \mid \mathrm{do}(x))$ (mass of $\{u : f_Y(x,u) = y\}$) are masses of unions of cells, so they agree with $\mathcal{M}_2$. ■

Hence, with $\mathcal{M}$ now the canonical models,

$$a(y;x) = \min_{M\in\mathcal{M}} P_M(y \mid \mathrm{do}(x)), \quad b(y;x) = \max_{M\in\mathcal{M}} P_M(y \mid \mathrm{do}(x)) \quad \text{s.t. } P_M(x, y \mid z) = P(x, y \mid z),$$

a linear program over $\mathbf{q} = (q_{j,k}) \in \mathbb{R}^{16}$.

5.1.3.4 LP formulation

Observational constraints. Each $p_{yx.z} = P(Y = y, X = x \mid Z = z)$ is linear in $\mathbf{q}$, e.g. ($Z = 1$, $X = 0$ needs $R_X \in \{0, 2\}$; $Y = 1$ at $X = 0$ needs $R_Y \in \{2, 3\}$)

$$\begin{aligned} p_{10.1} = P(Y = 1, X = 0 \mid Z = 1) &= P(\text{never-taker, hurt}) + P(\text{never-taker, always-recover}) \\ &\quad + P(\text{defier, hurt}) + P(\text{defier, always-recover}) \\ &= q_{0,2} + q_{0,3} + q_{2,2} + q_{2,3}. \end{aligned}$$

In full:

$$\begin{aligned} p_{00.0} &= q_{0,0} + q_{0,1} + q_{1,0} + q_{1,1}, & p_{00.1} &= q_{0,0} + q_{0,1} + q_{2,0} + q_{2,1}, \\ p_{01.0} &= q_{2,0} + q_{2,2} + q_{3,0} + q_{3,2}, & p_{01.1} &= q_{1,0} + q_{1,2} + q_{3,0} + q_{3,2}, \\ p_{10.0} &= q_{0,2} + q_{0,3} + q_{1,2} + q_{1,3}, & p_{10.1} &= q_{0,2} + q_{0,3} + q_{2,2} + q_{2,3}, \\ p_{11.0} &= q_{2,1} + q_{2,3} + q_{3,1} + q_{3,3}, & p_{11.1} &= q_{1,1} + q_{1,3} + q_{3,1} + q_{3,3}. \end{aligned}$$

Causal effect. Under $\mathrm{do}(X = 0)$, $Y = 1$ iff $R_Y \in \{\text{hurt, always-recover}\}$:

$$P(Y = 1 \mid \mathrm{do}(X = 0)) = P(R_Y = 2) + P(R_Y = 3) = \textstyle\sum_j (q_{j,2} + q_{j,3}),$$

$$P(Y = 1 \mid \mathrm{do}(X = 1)) = \textstyle\sum_j (q_{j,1} + q_{j,3}).$$

Optimization. $P(Y = 1 \mid \mathrm{do}(X = 0)) \in [a_{10}, b_{10}]$ with

$$a_{10} = \min_{\mathbf{q}} \sum_{j=0}^{3} (q_{j,2} + q_{j,3}), \qquad b_{10} = \max_{\mathbf{q}} \sum_{j=0}^{3} (q_{j,2} + q_{j,3})$$

subject to $\sum_{j,k} q_{j,k} = 1$, $q_{j,k} \geq 0$ and the eight equalities above.

5.1.3.5 Closed-form solution

Each bound is a linear combination of the $p_{yx.z}$ whose gap from the objective is non-negative:

$$P(Y = 1 \mid \mathrm{do}(X = 0)) \;\geq\; \max \begin{cases} p_{10.0} \\ p_{10.1} \\ p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1} \\ p_{01.0} + p_{10.0} - p_{00.1} - p_{01.1} \end{cases}$$

$$P(Y = 1 \mid \mathrm{do}(X = 0)) \;\leq\; \min \begin{cases} 1 - p_{00.0} \\ 1 - p_{00.1} \\ p_{10.0} + p_{11.0} + p_{01.1} + p_{10.1} \\ p_{01.0} + p_{10.0} + p_{10.1} + p_{11.1} \end{cases}$$

The first two entries are the natural bounds; the others tighten them.

Validity of $p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1}$. Substituting,

$$p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1} = q_{0,2} + q_{0,3} + q_{1,2} + q_{2,3} - q_{0,0} - q_{0,1} - q_{2,0} - q_{1,1},$$

$$\sum_j (q_{j,2} + q_{j,3}) - \big(p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1}\big) = q_{2,2} + q_{3,2} + q_{1,3} + q_{3,3} + q_{0,0} + q_{0,1} + q_{2,0} + q_{1,1} \;\geq 0.$$

Every closed-form bound is certified this way (LP duality: the coefficients are a feasible dual solution).

5.1.4 Back-door with partially observed covariates

Motivation. Bound an effect when the admissible set is partly measured but the missing part's marginal is known.

Let $\{W, U\}$ be back-door admissible ($W, U \to X$, $W, U \to Y$, W–U related, $X \to Y$), W observed, U not:

$$P(y \mid \mathrm{do}(x)) = \sum_{w,u} P(y \mid x, w, u) P(w, u) = \sum_{w,u} \frac{P(x, y, w, u) P(w, u)}{P(x, w, u)}.$$

Given $P(X, Y, W)$ and $P(U)$, the natural bound is $P(x, y) \le P(y \mid \mathrm{do}(x)) \le 1 - P(x, y')$ ($y' = 1 - y$, $x' = 1 - x$). Li and Pearl (2022) tighten it:

$$a(y; x) = \min_{a_{w,u}, b_{w,u}, c_{w,u}} \sum_{w,u} \frac{a_{w,u}\, b_{w,u}}{c_{w,u}}, \qquad b(y; x) = \max_{a_{w,u}, b_{w,u}, c_{w,u}} \sum_{w,u} \frac{a_{w,u}\, b_{w,u}}{c_{w,u}},$$

with $a_{w,u}, b_{w,u}, c_{w,u}$ the unknown $P(x, y, w, u)$, $P(w, u)$, $P(x, w, u)$, subject to, for all $w \in \mathfrak{X}_W$, $u \in \mathfrak{X}_U$,

$$\begin{gathered}
\textstyle\sum_u a_{w,u} = P(x, y, w), \qquad \sum_u b_{w,u} = P(w), \qquad \sum_u c_{w,u} = P(x, w), \\
b_{w,u} \ge c_{w,u} \ge a_{w,u} \quad (\text{since } P(x, y, w, u) \le P(x, w, u) \le P(w, u)), \\
\max\{0, P(x, y, w) + P(u) - 1\} \le a_{w,u} \le \min\{P(x, y, w), P(u)\}, \\
\max\{0, P(w) + P(u) - 1\} \le b_{w,u} \le \min\{P(w), P(u)\}, \\
\max\{0, P(x, w) + P(u) - 1\} \le c_{w,u} \le \min\{P(x, w), P(u)\}
\end{gathered}$$

(the last three lines are Fréchet bounds; $a_{w,u}$ alone suffices as parametrization).

5.2 Soft interventions

Motivation. Realistic policies install conditional or stochastic rules instead of constants, and the do-calculus extends to them.

5.2.1 Motivation

Motivation. Atomic $\mathrm{do}(x)$ is often not an implementable policy.

A tutoring program. Previous GPA W, motivation Z (low/high), tutoring X, final GPA Y: $W \to Z \to X \to Y$, $W \to Y$, $Z \to Y$.

ML can predict $P(y \mid w, z, x)$ well, but only in the current *natural regime*; decisions require predictions under an unrealised regime.

What we know so far. From $P(W, Z, X, Y)$ and $\mathcal{G}$,

$$\begin{aligned}
P(y \mid \mathrm{do}(x)) &= \sum_z P(y \mid \mathrm{do}(x), z) P(z \mid \mathrm{do}(x)) && \text{condition on } Z \\
&= \sum_z P(y \mid \mathrm{do}(x), z) P(z) && \text{Rule 3: } (Z \perp_d X)_{\mathcal{G}_{\overline{X}}} \\
&= \sum_z P(y \mid x, z) P(z) && \text{Rule 2: } (Y \perp_d X \mid Z)_{\mathcal{G}_{\underline{X}}}.
\end{aligned}$$

If $P(Y = 1 \mid \mathrm{do}(X = 1)) > P(Y = 1)$, $\mathrm{do}(X = 1)$ helps, but it means mandatory tutoring for every student.

Implementing do-interventions. Resources may be insufficient and compliance not guaranteed, so we need realizable interventions:

Do-like intervention	Realistic intervention
make sure no one smokes	reduce tobacco consumption to 20% of current
provide treatment to all patients	treat iff a patient is in critical condition
move a robotic arm to coordinate (x, y, z)	move it to (x, y, z) with a small error
mark all applicants as males	mark all applicants as males on paper

5.2.2 Canonical types of regimes

Motivation. Classify interventions by the mechanism they install.

- **Idle/natural:** $X = f_X(\mathbf{pa}_X, u_X)$.
- **Hard/atomic:** $X = x$, $\text{do}(X = x)$: every student is tutored.
- **Conditional:** $X = g(\mathbf{w})$ for observables $\mathbf{W}$: tutoring iff low GPA.
- **Stochastic:** $X \sim P^*(x \mid \mathbf{w})$: low-GPA students enter a raffle for 80% of the spots, other interested students for 20%.

Write σ_X for the intervention and $P(\cdot; \sigma_X)$ for the induced distribution.

Changes in regime. Data come from the natural regime with graph $\mathcal{G}$. The new regime's graph $\mathcal{G}_{\sigma_X}$ has a regime indicator (small square σ_X) pointing into the intervened variables, whose incoming edges follow the new mechanism.

5.2.3 Reduction to known machinery

Motivation. When do soft-intervention effects follow from hard-intervention results?

A reducible example. Let $P(X = 1 \mid Z = 1) = 0.3$ and double it, σ_X: $P^*(X = 1 \mid Z = 1) = 0.6$; X's parents stay $\{Z\}$. Then

$$\begin{aligned} P(y; \sigma_X) &= \sum_z P(y \mid z; \sigma_X) P(z; \sigma_X) = \sum_z P(y \mid z; \sigma_X) P(z) \\ &= \sum_{z,x} P(y \mid z, x; \sigma_X) P(x \mid z; \sigma_X) P(z) = \sum_{z,x} P(y \mid z, x; \sigma_X) P^*(x \mid z) P(z) \\ &= \sum_{z,x} P(y \mid z, x) P^*(x \mid z) P(z), \end{aligned}$$

since Z is unaffected, X follows $P^*(x \mid z)$, and Y given Z, X responds as before. Is a do-calculus-like derivation always enough?

An intentionally wrong derivation. Treatment X_1; a physician observes Z and picks X_2; survival Y. Graph: $X_1 \to Z$, $X_1 \to X_2$, $X_1 \to Y$, $Z \to X_2$, $X_2 \to Y$, $X_1 \leftrightarrow Z$, $Z \leftrightarrow Y$; policy $\sigma_{\mathbf{X}} = \{X_1 = x_1, X_2 = g(x_1, z)\}$. One might write

$$P(y \mid \text{do}(x_1), \text{do}(X_2 = g(x_1, z))) = P(y \mid x_1, \text{do}(X_2 = g(x_1, z))),$$

citing Rule 2 with $(Y \perp_d X_1 \mid X_2)$ in $\mathcal{G}_{\overline{X_2}\underline{X_1}}$. Wrong: edges into X_2 stay active, so the check must use $\mathcal{G}_{\sigma_{X_2}\underline{X_1}}$, where $X_1 \leftrightarrow Z \to X_2 \to Y$ is open.

Where do-calculus intuition breaks. Do-interventions cut all dependence on parents; soft ones may keep some, change their distribution, or add new ones.

5.2.4 σ-calculus

Motivation. Inference rules for soft interventions consult both pre- and post-interventional graphs.

Theorem 5.5 (σ-calculus, simplified; Correa and Bareinboim 2020)

For disjoint $\mathbf{X}, \mathbf{Y}, \mathbf{Z} \subseteq \mathbf{V}$ and disjoint $\mathbf{T}, \mathbf{W} \subseteq \mathbf{V} \setminus (\mathbf{Z} \cup \mathbf{Y})$:

Rule 1 (insertion/deletion of observations) $P(\mathbf{y} \mid \mathbf{w}, \mathbf{t}; \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{T} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}}$.

Rule 2 (change of regimes under observation) $P(\mathbf{y} \mid \mathbf{w}, \mathbf{x}; \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}, \mathbf{x})$ if $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\underline{\mathbf{X}}}$ and in $\mathcal{G}_{\underline{\mathbf{X}}}$.

Rule 3 (change of regimes without observation) $P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w})$ if $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\overline{\mathbf{X(W)}}}$ and in $\mathcal{G}_{\overline{\mathbf{X(W)}}}$.

Proof [Supplement]. As for do-calculus (Chapter 3), with the regime indicator selecting between the natural mechanism and $\sigma_{\mathbf{X}}$. **Rule 1**: the post-regime distribution is Markov to $\mathcal{G}_{\sigma_{\mathbf{X}}}$. **Rules 2 and 3**: add a switch $F_X \to X$ with values σ and idle; as in do-calculus, $F_X \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W}, \mathbf{X}$ (Rule 2) or $F_X \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W}$ (Rule 3) follows from the separation with outgoing (resp. incoming) edges of $\mathbf{X}$ removed. A soft regime may add parents of $\mathbf{X}$, so the separation must hold in the graph of each regime, hence both $\mathcal{G}_{\sigma_{\mathbf{X}}\cdots}$ and $\mathcal{G}_{\cdots}$. ■

Full version. With outer regime $\sigma_{\mathbf{X}}$ and a change on $\mathbf{Z}$ from $\sigma_{\mathbf{Z}}$ to $\sigma'_{\mathbf{Z}}$:

- Rule 1: $P(\mathbf{y} \mid \mathbf{w}, \mathbf{t}; \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{T} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}}$;
- Rule 2: $P(\mathbf{y} \mid \mathbf{w}, \mathbf{z}; \sigma_{\mathbf{Z}}, \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}, \mathbf{z}; \sigma'_{\mathbf{Z}}, \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{Z} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma_{\mathbf{Z}}\underline{\mathbf{Z}}}$ and $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma'_{\mathbf{Z}}\underline{\mathbf{Z}}}$;
- Rule 3: $P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{Z}}, \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}; \sigma'_{\mathbf{Z}}, \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{Z} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma_{\mathbf{Z}}\overline{\mathbf{Z(W)}}}$ and $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma'_{\mathbf{Z}}\overline{\mathbf{Z(W)}}}$.

5.2.4.1 Comparison with do-calculus

Remark. Under $\mathrm{do}(x)$, $P(y \mid \mathrm{do}(x), w) = P(y \mid \mathrm{do}(x), w, x)$; under a non-atomic intervention, in general $P(y; \sigma_X) \neq P(y \mid x; \sigma_X)$ for $X \to Y$.

	do-calculus	σ-calculus
Rule 1	$P(y \mid \mathrm{do}(x), w, t) = P(y \mid \mathrm{do}(x), w)$ if $(Y \perp_d T \mid W, X)_{\mathcal{G}_{\overline{X}}}$	$P(y \mid w, t; \sigma_X) = P(y \mid w; \sigma_X)$ if $(Y \perp_d T \mid W)_{\mathcal{G}_{\sigma_X}}$
Rule 2	$P(y \mid \mathrm{do}(x), w) = P(y \mid x, w)$ if $(Y \perp_d X \mid W)_{\mathcal{G}_{\underline{X}}}$	$P(y \mid x, w; \sigma_X) = P(y \mid x, w)$ if $(Y \perp_d X \mid W)$ in $\mathcal{G}_{\sigma_X \underline{X}}$ and $\mathcal{G}_{\underline{X}}$
Rule 3	$P(y \mid \mathrm{do}(x), w) = P(y \mid w)$ if $(Y \perp_d X \mid W)_{\mathcal{G}_{\overline{X(W)}}}$	$P(y \mid w; \sigma_X) = P(y \mid w)$ if $(Y \perp_d X \mid W)$ in $\mathcal{G}_{\sigma_X \overline{X(W)}}$ and $\mathcal{G}_{\overline{X(W)}}$

Rule 1's graph depends on the intervention; Rules 2 and 3 need both graphs.

Revisiting Rule 2. For the sequential treatment,

$$P(y \mid \mathrm{do}(x_1), \mathrm{do}(X_2 = g(x_1, z))) = P(y \mid x_1, \mathrm{do}(x_1), \mathrm{do}(X_2 = g(x_1, z))) \\ = P(y \mid x_1; \sigma_{X_1, X_2}) \stackrel{?}{=} P(y \mid x_1; \sigma_{X_2}).$$

Rule 2 needs no back-door path $X_1 \to Y$ both when $X_2 = g$ ignores Z and when X_2 listens to X_1, Z; only the first holds, so the rule does not apply.

5.2.4.2 Another intervention

Conditional policy: only low-GPA students ($W = 0$) are tutored, $P^*(X = 1 \mid W = 0) = 1$, $P^*(X = 1 \mid W = 1) = 0$. In $\mathcal{G}_{\sigma_X}$, X depends on W (not Z):

$$P(y; \sigma_X) = \sum_{w,z,x} P(y \mid x, w, z; \sigma_X) P(x \mid w, z; \sigma_X) P(w, z; \sigma_X)$$

$$= \sum_{w,z,x} P(y \mid x, w, z; \sigma_X) P(x \mid w; \sigma_X) P(w, z; \sigma_X) \qquad \text{Rule 1}$$

$$= \sum_{w,z,x} P(y \mid x, w, z) P(x \mid w; \sigma_X) P(w, z; \sigma_X) \qquad \text{Rule 2}$$

$$= \sum_{w,z,x} P(y \mid x, w, z) P^*(x \mid w) P(w, z) \qquad \text{Rule 3.}$$

5.2.5 C-factors and soft interventions

Motivation. Identify soft-intervention effects systematically with c-factors.

For $\mathcal{G}$: $Z \to X$, $Z \to Y$, $X \to Y$,

$$P(z, x, y) = \sum_{u_Z} P(z \mid u_Z) P(u_Z) \sum_{u_X} P(x \mid z, u_X) P(u_X) \sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y) = Q[Z]\, Q[X]\, Q[Y].$$

Under $\mathcal{M}' = \mathcal{M}_{\sigma_X}$ (diagram $\mathcal{G}' = \mathcal{G}_{\sigma_X}$) only X's mechanism changes:

$$P(z, x, y; \sigma_X) = \sum_{u_Z} P(z \mid u_Z) P(u_Z) \sum_{u'_X} P'(x \mid z, u'_X) P(u'_X) \sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y)$$

$$= Q[Z]\, Q[X; \sigma_X]\, Q[Y].$$

In general $P(\mathbf{y}; \sigma_{\mathbf{X}}) = \sum_{\mathbf{d} \setminus \mathbf{y}} \prod_{\mathbf{D}_i \in \mathcal{C}(\mathbf{D})} Q[\mathbf{D}_i]$ over the c-components of $\mathcal{G}_{\sigma_{\mathbf{X}}}$ restricted to the relevant ancestors; $Q[\mathbf{X}; \sigma_{\mathbf{X}}]$ is given by $\sigma_{\mathbf{X}}$, the other factors are identified from $P(\mathbf{V})$, e.g. by IDENTIFY (Chapter 3).

Example 5.2 (Systematic identification 1)

$\mathcal{G}$: $W \to Z$, $X \to Z \to Y$, $X \leftrightarrow Y$; $\sigma_X = P^*(X \mid W)$, so $\mathcal{G}_{\sigma_X}$ gains $W \to X$ and loses $X \leftrightarrow Y$. C-components: $\mathcal{G}$: $\mathbf{C}_1 = \{X, Y\}$, $\mathbf{C}_2 = \{W\}$, $\mathbf{C}_3 = \{Z\}$; $\mathcal{G}_{\sigma_X}$: $\mathbf{D}_1 = \{Y\}$, $\mathbf{D}_2 = \{W\}$, $\mathbf{D}_3 = \{Z\}$, $\mathbf{D}_4 = \{X\}$, with $Q[\mathbf{D}_4] = Q[X; \sigma_X]$ given. IDENTIFY$(\{Y\}, \{X, Y\}, Q[T], \mathcal{G})$: $A = An(\{Y\})_{\mathcal{G}[\{X,Y\}]} = \{Y\} = C$, so $Q[Y] = \sum_x Q[X, Y]$; $\{W\}$ and $\{Z\}$ return immediately. Hence

$$P(y; \sigma_X) = \sum_{w,x,z} Q[X; \sigma_X]\, Q[Z]\, Q[W]\, Q[Y] = \sum_{w,x,z} Q[X; \sigma_X]\, Q[Z]\, Q[W] \sum_{x'} Q[X, Y]$$

$$= \sum_{w,x,z} P^*(x \mid w)\, P(z \mid x, w)\, P(w) \sum_{x'} P(x')\, P(y \mid x', z, w).$$

Example 5.3 (Systematic identification 2: failure)

$\mathcal{G}$: $W \to Z \to X \to Y$, $W \leftrightarrow X$, $W \leftrightarrow Y$; $\sigma_X = P^*(X \mid Z)$. C-components: $\mathcal{G}$: $\mathbf{C}_1 = \{W, X, Y\}$, $\mathbf{C}_2 = \{Z\}$; $\mathcal{G}_{\sigma_X}$: $\mathbf{D}_1 = \{Y, W\}$, $\mathbf{D}_2 = \{X\}$, $\mathbf{D}_3 = \{Z\}$. IDENTIFY$(\{W, Y\}, \{W, X, Y\}, Q[T], \mathcal{G})$: $A = An(\{W, Y\})_{\mathcal{G}[\{W,X,Y\}]} = \{W, X, Y\} \neq C$ and $A = T$, so FAIL. (Recall $Q[\mathbf{C}_j] = \prod_{V_i \in \mathbf{C}_j} Q[\mathbf{V}^{\leq i}]/Q[\mathbf{V}^{\leq i-1}]$, $Q[\mathbf{V}^{\leq i}] = \sum_{\mathbf{v}^{>i}} Q[\mathbf{V}]$.)

5.2.6 Summary

- Implementable policies are often conditional or stochastic, not atomic.
- σ-calculus rules consult pre- and post-interventional graphs.
- C-factor procedures identify soft-intervention effects systematically.